

TEST REPORT INSTRUCTIONS

早期 | 全局 | 动态 | 精准

血液循环肿瘤DNA甲基化

Blood circulation tumor DNA methylation



CONTACT

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01

本检测方法基于华微智检MePlex技术平台,对血液或其他体液存在微量的肿瘤来源DNA(ctDNA)标记信号进行检测,实现对肿瘤的早期风险评估或肿瘤治疗监测;

This detection method is based on the HuaWeiBio, MePlex technology platform. It detects marker signals of tumor-derived DNA (ctDNA) present in trace amounts in blood or other body fluids, enabling early tumor risk assessment or tumor treatment monitoring.

02

肿瘤易受体内用药等微环境变化影响导致短暂休眠或加速进展情况,通过连续检测对比可以有效反应肿瘤进展状态,请定期检测关注信号变化趋势;

Tumors can undergo temporary dormancy or accelerated progression due to microenvironmental changes such as medication. Continuous monitoring can effectively reflect the tumor's status, so it is advised to track these changes regularly.

03

虽然本检测方法经研发升级优化达到了行业领先的灵敏度和特异性,由于个体差异环境因素、用药等影响,仍有一定假阳性和假阴性的可能;

Although MePlex has achieved industry-leading sensitivity and specificity through research and development improvements, factors like individual variability and environmental influences may still result in false positives or false negatives.

04

本次检测结果仅代表您血液采集时的生理疾病状态,同一人不同时期的检测结果可能存在差异,因肿瘤发生发展周期较长,建议定期检测进行健康管理;

The results represent your physiological state at the time of blood collection only, with possible variations occurring over different periods due to the lengthy cycle of tumor growth and development. Regular testing is recommended for ongoing health management.

05

本检测通过血液游离DNA甲基化肿瘤标记物检测是否存在肿瘤来源DNA片段,并非直接检测潜在肿瘤的原发部位,因此检测结果并非最终病理确定诊断,不代表临床决策意见;

This test detects tumor markers through free DNA in the blood to indicate the presence of tumor-derived DNA segments, but it does not directly pinpoint the primary location of a potential tumor, hence the results are not a definitive pathological diagnosis.

06

随着科技的进步和研发技术的优化,相关检测方法和原理可能会有更新;

With advances in technology and refinement of research techniques, updates to the methods and principles of testing may occur.

07

本次检验结果仅对当次标本负责,结果供临床研究参考,如有疑问请在收到结果7个工作日内与我们联系;

The responsibility of this test result is limited to the specimen currently tested, and it is intended solely for clinical research purposes. If there are any questions, please request a retest within seven working days from the report date; requests will not be accepted after this period.

08

我司承诺检测实验流程严格遵守国家相关法律法规及实验室管理规范,对受检者个人信息和遗传信息严格保密,不会对其他任何第三方泄露受检者的隐私信息。

Our company assures that the testing process strictly adheres to national legal regulations and laboratory management standards, maintaining strict confidentiality regarding personal and genetic information of the tested individuals, and ensuring no disclosure of such private information to any third parties.

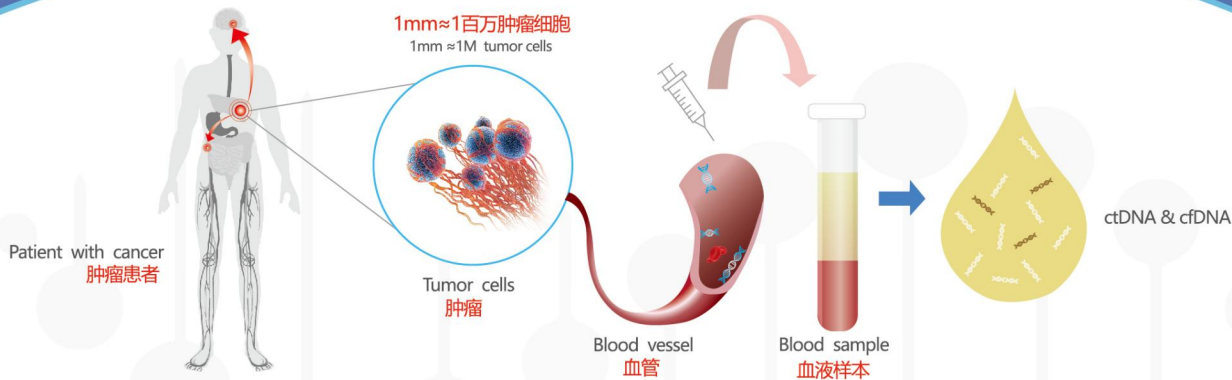
◆ CIRCULATING TUMOR DNA (ctDNA) 循环肿瘤DNA

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循环肿瘤DNA(ctDNA)是指肿瘤细胞凋亡或坏死时释放到血液中的碎片化DNA，其序列特征与原始肿瘤的基因组相符合。仅1~3毫米大小的肿瘤可包含100万~2700万个肿瘤细胞，因此，肿瘤进展过程中大量凋亡肿瘤细胞DNA片段会进入到血液中，可以从患者血液样本中检测出ctDNA。

相比1~3毫米的肿瘤很难通过影像学技术发现，因此ctDNA的检测敏感性要高于传统影像学技术。

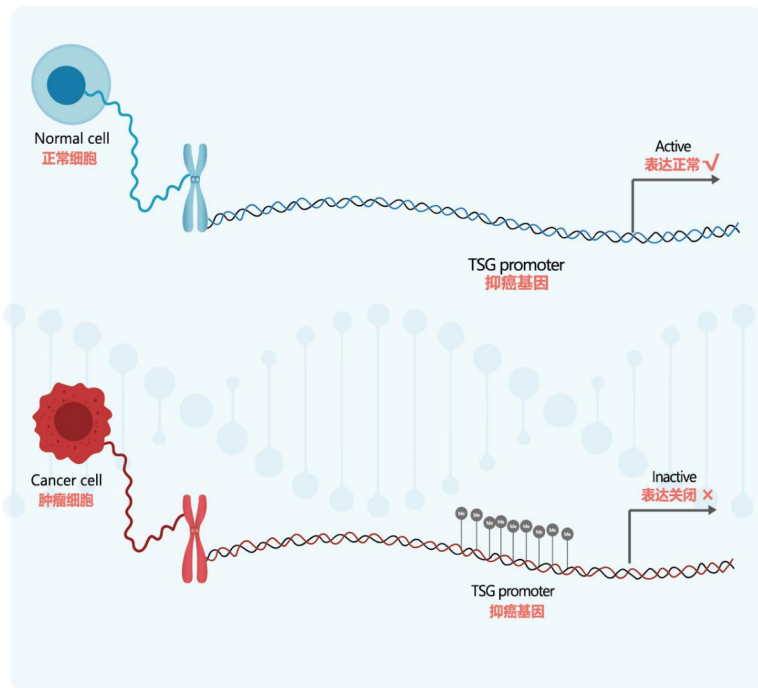
Circulating tumor DNA (ctDNA) consists of fragmented DNA from apoptotic or necrotic tumor cells that enter the bloodstream, matching the genomic sequence of the original tumor. Even a tumor as small as 1-3 mm can release significant amounts of ctDNA due to millions of dying cells, making it detectable in blood samples. This feature renders ctDNA detection more sensitive than imaging for identifying small tumors.



DNA甲基化是一种不改变DNA序列而能够调控基因表达的遗传修饰方式。DNA甲基化异常在肿瘤的发生与进展中扮演关键角色。**DNA甲基化的变化发生在肿瘤形成阶段，是目前发现最理想的肿瘤标志物。**通过分析血液或组织样本中的甲基化模式，医生可以实施肿瘤筛查和诊断，从而推动疾病的早期发现和及时治疗，进而提升治愈率。



DNA methylation, a genetic modification that regulates gene expression without altering the DNA sequence itself, plays a pivotal role in cancer development and progression. Changes in methylation patterns, occurring at the onset of tumorigenesis, are among the most promising tumor markers. By analyzing the methylation profiles in blood or tissue samples, doctors can screen and diagnose cancers, facilitating early detection and treatment, thereby improving survival rates.



ctDNA甲基化技术在肿瘤的早期检测、治疗评价及微小残留病灶 (MRD) 的监测方面已经得到了专业共识, 标志着医学领域的一大科技进展和变革。这项技术在结肠癌、肺癌、乳腺癌等高发肿瘤的精确治疗上展示了其关键价值和临床支持作用。

The ctDNA methylation technology has gained professional consensus in the early detection of tumors, treatment evaluation, and monitoring of minimal residual disease (MRD), marking a major technological advancement and transformation in the medical field. This technology has demonstrated its critical value and clinical support role in the precise treatment of high-incidence tumors such as colon cancer, lung cancer, and breast cancer.

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专家共识

肿瘤 DNA 甲基化标志物检测及临床应用专家共识 (2024 版)

中国抗癌协会肿瘤标志专业委员会

【摘要】随着生物医学的发展, DNA 甲基化(DNA methylation)标志物在癌症诊断、治疗选择及预后评估中的重要性日益凸显。本专家共识旨在全面梳理 DNA 甲基化标志物的检测技术、临床应用及其在肿瘤管理中的潜力。此外, 本共识将围绕 DNA 甲基化标志物的定义、临床意义、检测规范、数据处理及其在肿瘤的筛查、辅助诊断、伴随诊断及复发监测中的应用, 结合最新研究进展和实际临床经验, 提出一系列关于 DNA 甲基化标志物检测及应用的共识推荐, 旨在提升临床医技人员对这一新兴标志物的认识, 规范检测流程, 促进其在肿瘤诊疗全流程中的应用, 从而为患者提供更加精准有效的治疗方案。

【关键词】 DNA 甲基化; 生物标志物; 肿瘤早期诊断; 甲基化检测技术

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通过提升肿瘤早期诊断的准确率, ctDNA甲基化为实现个体化治疗方案提供了可靠的分子依据, 进一步促进了治疗效果的有效评估和疾病复发的及时监测。

这一进展不仅极大地推动了肿瘤管理和治疗的发展, 而且展现出了深远的临床实用价值和潜力。

By improving the accuracy of early tumor diagnosis, ctDNA methylation provides a reliable molecular basis for achieving individualized treatment plans, further facilitating the effective assessment of treatment outcomes and the timely monitoring of disease recurrence. This advancement not only significantly promotes the development of tumor management and treatment, but also demonstrates profound clinical practical value and potential.



肿瘤早检

通过ctDNA甲基化检测可以比影像学提前一年左右发现肿瘤存在。此外，分析个体的DNA甲基化模式可以揭示受检者潜在的细胞功能异常，用于评估亚健康状态。

Early Cancer Detection

Testing for ctDNA methylation can uncover tumors approximately a year earlier than imaging. Additionally, individual DNA methylation patterns may reveal potential cellular dysfunction for assessing suboptimal health states.



疗效评估

ctDNA甲基化检测能够反映治疗过程中肿瘤负荷。通过在治疗过程中连续检测可以监测肿瘤信号溢出的动态变化情况，判断药物治疗的效果，有助于指导后续的治疗方案。

Treatment Efficacy

ctDNA methylation levels can indicate tumor burden during treatment. Continuous monitoring allows for the evaluation of dynamic changes in tumor-derived signals, assessing the effectiveness of drugs and guiding future therapies.



手术残留

肿瘤复发的主要原因是手术切除后仍然存在微小残留肿瘤病灶，ctDNA甲基化是分子水平的检测相比影像学技术可以发现更微弱残留肿瘤病灶的存在，有助于指导后续的治疗方案。

Surgical Residue

Molecular detection of ctDNA methylation can identify minute residual tumors post-surgery, which are often missed by imaging, aiding in planning further treatments.

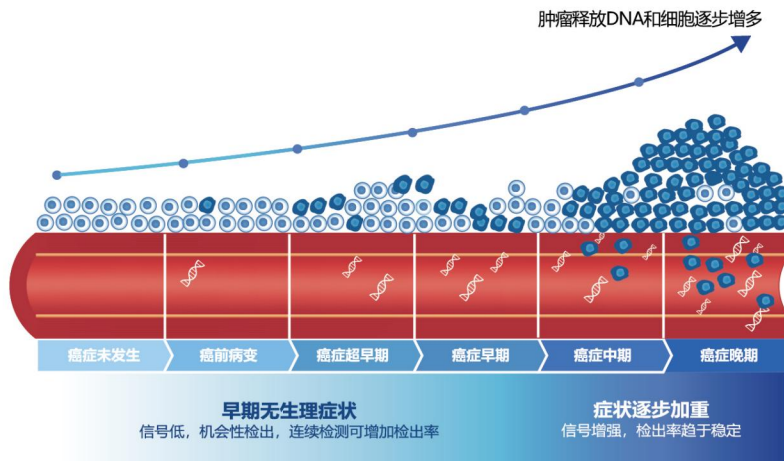
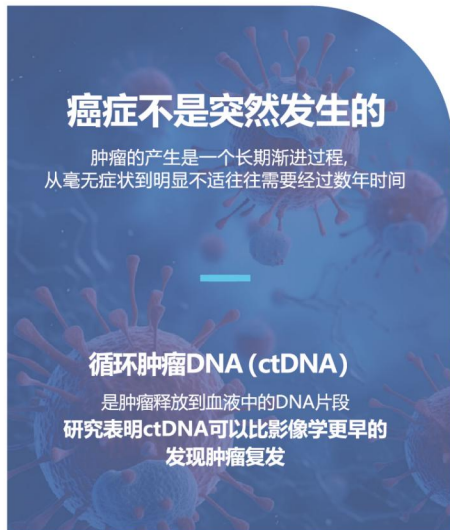


复发监测

ctDNA 检测是一种非侵入性的液体活检方法，通过分析血液中的 ctDNA 甲基化信号，可以提前 2-12 个月发现肿瘤复发信号，提示及早介入治疗，可有效改善患者生存率。

Relapse Monitoring

ctDNA testing—a non-invasive liquid biopsy—detects relapse signals 2-12 months in advance by analyzing ctDNA methylation in the blood, prompting early therapeutic intervention and potentially improving patient survival rates.



医学科技进步让肿瘤可检出口逐渐前移

在肿瘤形成早期、影像还看不到病灶的阶段，体内已出现微量肿瘤相关信号（如ctDNA甲基化），是早诊早治的关键时机。然而肿瘤信号检测存在窗口期，单次检测可能受身体状态、用药等因素影响出现偏差，多次检测观察信号的动态变化更有参考意义，可区分良性波动与真实风险，为早期预警和诊疗提供可靠依据。

In early tumor development, when lesions are not visible by imaging, trace tumor-related signals (e.g., ctDNA methylation) already exist, offering a critical window for early diagnosis and treatment. However, single tests may be affected by physical condition or medication, leading to bias. Multiple tests and dynamic signal monitoring are more reliable to distinguish benign fluctuations from real risks, supporting early warning and clinical decision-making.

◆ GENE BIOMARKERS 基因点位

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华微智检对上万例肿瘤样本大数据分析，优选抑癌基因和原癌基因DNA甲基化标记库MEBANK。通过对以下基因标记的组合，即可构建10+多分子靶标的分子标记组合，用以更精准的识别或监测不同类型的肿瘤。

Leveraging big data from thousands of tumor samples, HuaWeiBio has curated MeBank, a collection of methylation markers for tumor suppressor and oncogenes. A panel of over 10 multimolecular targets is constructed using these gene markers to precisely identify and monitor various cancer types.

基因标记库部分点位示例 | Partial Display of Methylation Biomarker Bank

BRCA1, 维持基因组稳定性; Maintains genomic stability	RB1, 调控细胞周期; Regulates cell cycle
p53 (TP53), 诱导凋亡和DNA修复; Induces apoptosis and DNA repair	PTEN, 抑制PI3K/AKT通路; Inhibits PI3K/AKT pathway
APC, Wnt通路信号转导抑制; Inhibits Wnt signaling pathway	VHL, 调节细胞对氧感应; Regulates cellular response to oxygen
MGMT, 修复O6- ** 鸟嘌呤; Repairs O6-methylguanine	CDH1, 维持细胞间黏附; Maintains cell adhesion
CDKN2A (p16), 抑制细胞周期进程; Inhibits cell cycle progression	MLH1, 参与错配修复机制; Involved in mismatch repair
MSH2, 错配修复蛋白复合体组分; Component of mismatch repair complex	BRCA2, 维持基因组稳定性; Maintains genomic stability
FOXO1, 调节细胞凋亡和代谢; Regulates apoptosis and metabolism	SCGB3A1, 肺部肿瘤抑制; Lung tumor suppression
ESR1, 调节细胞增殖和分化; Regulates cell proliferation and differentiation	FHIT, 涉及DNA损伤应答; Involved in DNA damage response
TCF21, 细胞分化和肿瘤抑制; Cell differentiation and tumor suppression	GATA5, 参与心脏发育和表型; Involved in heart development and phenotype
SFRP1, 调节Wnt信号和抗肿瘤; Modulates Wnt signaling and anti-tumor	MYC, 促进细胞增殖; Promotes cell proliferation
KRAS, 细胞信号传导调节; Regulates cell signaling	BRAF, 信号传递通路中介; Mediates signaling pathways
PIK3CA, 激活AKT通路; Activates AKT pathway	BCL2, 抑制程序性细胞死亡; Inhibits programmed cell death
CCND1, 促进G1/S细胞周期过渡; Promotes G1/S cell cycle transition	ERBB2, 细胞生长和分化信号; Cell growth and differentiation signaling
JUN, 调控细胞增殖和分化; Regulates cell proliferation and differentiation	MET, 细胞增殖和存活信号; Cell proliferation and survival signaling
H-RAS, 调节多种细胞过程; Regulates multiple cellular processes	N-MYC, 细胞生长和代谢调节器; Cell growth and metabolism regulator

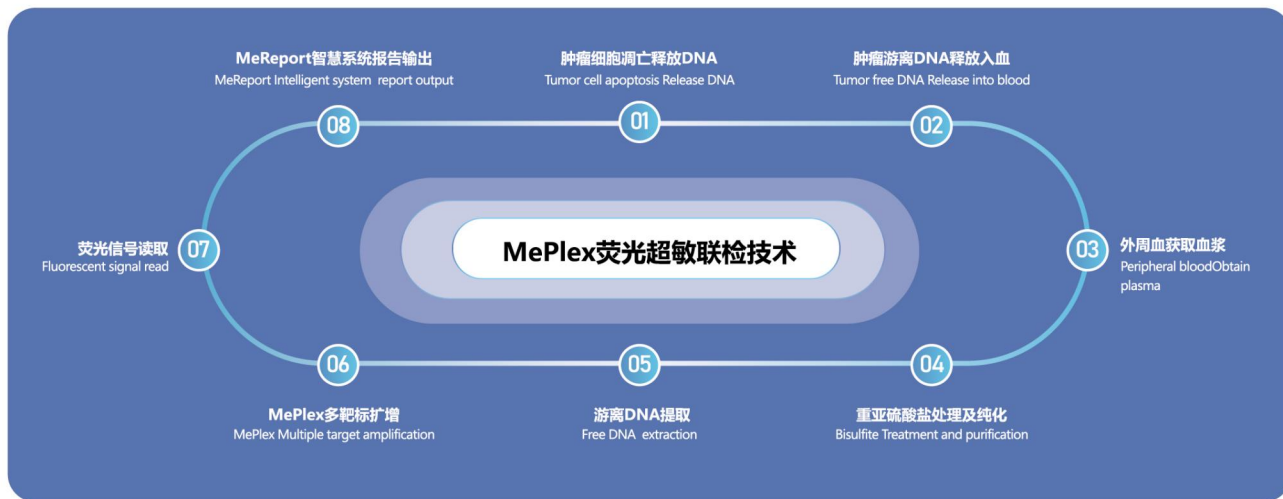
MePlex荧光超敏联检技术

MePlex创新技术有效解决痕量核酸多甲基化标记联合检测难题，具有灵敏度高、标记多、耗时短等技术优势。该项创新成果领跑国际同行并率先进入临床应用。

MePlex技术极大地增强了医疗专业人员在肿瘤识别、治疗效果评估与复发监测方面的能力。通过MePlex技术，传统依赖影像学等方法的限制被有效突破，它不仅提升了临床诊断的准确性，还满足了逐渐增加的临床需求，促进了对病理状态更深入的理解和诊断。

MePlex Fluorescence Assay

MePlex is an innovative technology that resolves the challenge of multi-marker labeling and detection of trace nucleic acids with high sensitivity, multiplexing capability, and efficiency. A leader in its field and a pioneer in clinical application, MePlex greatly enhances medical professionals' ability to identify tumors, assess treatment responses, and monitor recurrences. Overcoming limitations of traditional imaging methods, MePlex not only improves diagnostic precision but also meets the growing clinical demand, deepening the understanding and diagnosis of pathological conditions.



◆ FREQUENTLY ASKED QUESTIONS 常见问题

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Q1.非肿瘤受检者是否会检测到DNA甲基化阳性?

怀孕、外伤、糖尿病、关节炎、免疫性疾病、感染、炎症等病症会导致个别DNA甲基化检测靶点的甲基化现象，但一般这种非肿瘤性信号导致多个靶点联合检出阳性概率较低。

Can non-cancer patients test positive for DNA methylation?

While conditions such as pregnancy, injury, diabetes, arthritis, immunological disorders, infections, and inflammation can trigger methylation at certain DNA points, the likelihood of a positive result across multiple targets is low in noncancer cases.

Q3.我还需要做一些其他的附加检查吗?

由于肿瘤发病机制极其复杂，存在巨大的人群及肿瘤异质性，所有检查技术都有优势和弊端，检查结果存在一定的制约和限度。此外，临床医生在进行确定诊断时需要结合多个检测特征综合评估后得出结论。

Do I need additional tests?

Given cancer's complexity and heterogeneity, all screening technologies have their pros and cons and limitations. Diagnosis often requires a comprehensive assessment of multiple indicators by clinicians.

Q5.如何做好肿瘤检查?

积极预防和定期接受全面的检查是最好的办法。相反，过于相信自身的健康状况或因害怕面对检查结果（除非已经产生实质性症状）而不选择进行检查是极其不明智的。

How can I ensure effective cancer screening?

The best approach is proactive prevention and regular comprehensive checks. Ignoring health checks due to fear or overconfidence in one's health, until symptoms manifest, is not advisable.

Q7.本检测技术主要缺点?

并非直接检测患病部位而是通过血液进行检测，因此存在微量肿瘤来源DNA片段被人体约5000ML血液稀释导致没有采集到目标DNA的情况。

What are the drawbacks?

This blood-based test could miss capturing the target DNA if the minute tumor-derived fragments are diluted in approximately 5000ml of blood.

Q9.适合接受检测的人群?

从官方公布肿瘤发病统计数据来看，无论男女老少均有患癌几率。所以，所有人群都是检查的对象，特别是有家族史、现病史、亚健康状态或年龄超过45岁的人群更需要接受检测。

Who should get tested?

Cancer affects all demographics, so everyone should be screened, especially those with family or current medical histories, suboptimal health statuses, or aged above 45.

Q11.与通常的检查有什么不同?

影像学是通过整体肿瘤状态来评估，病理学是从细胞水平进行细胞形态分析，而本检测是通过导致肿瘤发病根源的分子水平来检测分析。

How is this different from usual assessments?

Whereas imaging assesses the overall state of a tumor and pathology looks at cell morphology, this test analyzes at a molecular level, identifying the root causes of cancer development.

Q2.如何正确看待检测结果?

对于肿瘤检查不要过于紧张或焦虑，如果对阳性结果过度的解读或不安反而会会影响身体健康状况以及健康管理决策，应该综合考虑近期自身身体状态或结合其他检测手段，正确对待检测结果并用积极的心态去面对。

How should I interpret my test results?

Avoid excessive worry over tumor screenings. Misinterpreting positive results can harm your health and influence medical decisions. Consider your overall health and other tests when reviewing results and maintain a positive attitude.

Q4.该检测结果是否可以用于病理诊断?

临床病理诊断是通过穿刺或手术方式直接获取病灶组织，可以直接用于确定诊断，该检测是通过间接的方式捕获微量肿瘤来源DNA片段，因此不用于确定性病理诊断。

Can these test results be used for pathological diagnosis?

Clinical pathological diagnosis involves direct tissue sampling, which is definitive. This test captures trace tumor-derived DNA fragments indirectly, hence it's not used for conclusive pathology.

Q6.本检测技术的主要优点?

几乎没有传统肿瘤检查的缺陷(禁食、事前准备、创伤、疼痛、辐射等)，可以比影像学提前检出肿瘤信号并且适用于肿瘤信号持续监测。

What are the main advantages of this test?

It largely overcomes traditional limitations like fasting, pre-test preparations, trauma, pain, and radiation, detecting tumor signals earlier than imaging and is applicable for continuous monitoring.

Q8.本检测技术有何制约?

在受检者存在怀孕、抗癌治疗、外科手术、糖尿病等基础疾病状态下，会导致良性病变DNA释放到血液中，可能造成假阴性或假阳性结果。

What might affect this test?

Pregnancy, cancer treatment, surgery, diabetes, and other underlying conditions could release benign DNA into the bloodstream, leading to false negatives or positives.

Q10.感冒对检测结果有影响吗?

个别靶点检测结果可能会高一些，但总体不影响结果判断。

Will having a cold affect the test?

Although certain targets may show higher results, overall interpretation remains unaffected.

Q12.日常生活习惯对检测结果有影响吗?

吸烟饮酒、饮食偏好、作息不规律、不良嗜好或药物滥用等会导致部分靶点阳性检出的情况。

Does lifestyle affect the results?

Smoking, drinking, dietary habits, irregular sleep, bad habits, or drug abuse can lead to positive results for some targets.

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关于华微智检

华微智检专注于肿瘤早筛和诊断技术，融合研发、生产服务等一体化运营。针对肿瘤高发趋势，公司推动健康管理理念，响应国策，自主创新研发适用多种肿瘤的DNA化超敏检测技术，解决微量血液中游离DNA化检测难题，达成领域突破，旨在实现肿瘤早期准确发现及病程管理，提升治疗效果，改善患者生活品质。

About HuaWeiBio

Specializing in early cancer screening and diagnostic technologies, HuaWei Intelligent Inspection integrates R&D with production services for seamless operations. Facing the rising trend of cancer, the company advances health management initiatives aligned with national policies, innovates autonomously in R&D, and develops highly sensitive DNA detection technologies suitable for various cancers. Their breakthrough in detecting trace amounts of free DNA in blood ensures early and accurate identification of cancer and disease progression management, thus enhancing treatment results and improving patients' quality of life.



关于禾硕医学

苏州禾硕医学检验实验室秉承 ISO15189 质量管理体系以及CAP认证标准，致力于实施严格的质量管理流程。实验室专注构建一流的基因检测技术体系，并推动标准化的新型临床检测项目有效转化，旨在为医疗从业者和患者群体提供精准、可靠的技术服务。我们的目标是提供给患者具有深远医疗价值的服务项目。

About HeShuo Medical

Adhering to ISO 15189 quality management systems and CAP standards, Suzhou Heshuo Medical Laboratory is committed to stringent quality control measures. The lab focuses on building a first-class genetic testing technology framework and promoting the standardization of novel clinical test projects for effective implementation, aiming to provide precise and reliable technical services to healthcare providers and patients. Our goal is to deliver impactful and medically valuable services to patients.



战略合作伙伴

华微智检与苏州禾硕医学检验实验室携手合作，携其在医疗行业内的专业能力与经验，共同推出了针对肿瘤的早期检查、诊断及治疗管理等一系列服务。通过这一合作，我们旨在优化患者的医疗体验，并提高治疗效果，最终改善患者的生活质量。

Company Strategic Partner

HuaWeiBio has forged a strategic partnership with Suzhou Heshuo Medical Laboratory, leveraging their expertise and experience in the healthcare industry to introduce a comprehensive suite of services including early detection, diagnosis, and treatment management for cancer. This collaboration seeks to enhance the patient care experience, improve treatment outcomes, and better patients' quality of life.





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